Name
Date
Course

Pre-Algebra Practice Test III - Version One

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show all work.
iii) Draw a rectangle around your answers.
iv) Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

Solve.

1) $2x = 6x - 20$

$-6x + 2x = 6x - 20 - 6x$

$-4x = -20$

$\frac{-4x}{-4} = \frac{-20}{-4}$

$\boxed{x = 5}$

2) $8 - 7y + 3 + y = 5 + 3y$

$11 - 6y = 5 + 3y$

$6y + 11 - 6y = 5 + 3y + 6y$

$11 = 5 + 9y$

$-5 + 11 = 5 + 9y - 5$

$6 = 9y$

$\frac{6}{9} = \frac{9y}{9}$

$\boxed{\frac{2}{3} = y}$

3) $2(3z - 5) + 1 = -(z + 1)$

$6z - 10 + 1 = -1(z + 1)$

$6z - 9 = -z - 1$

$7z = 8$

$\boxed{z = \frac{8}{7}}$

4) $-x + 1 = 3 - 5x$

$5x - x + 1 = 3 - 5x + 5x$

$4x + 1 = 3$

$-1 + 4x + 1 = 3 - 1$

$4x = 2$

$\frac{4x}{4} = \frac{2}{4}$

$\boxed{x = \frac{1}{2}}$

5) $3 = 4 + 2(x - 1) - 8$

$3 = 4 + 2x - 2 - 8$

$3 = -6 + 2x$

$9 = 2x$

$\boxed{\frac{9}{2} = x}$

6) $-3(2z - 3) + 1 = 4$

$-6z + 9 + 1 = 4$

$-6z + 10 = 4$

$-6z = -6$

$\boxed{z = 1}$

7) $-2 - 8x = 2 - x$

$8x - 2 - 8x = 2 - x + 8x$

$-2 = 2 + 7x$

$-2 - 2 = 2 + 7x - 2$

$-4 = 7x$

$\frac{-4}{7} = \frac{7x}{7}$

$\boxed{-\frac{4}{7} = x}$

Evaluate.

8) $4 - \frac{x}{2}$, $x = -8$

$4 - \left(\frac{-8}{2}\right)$

$4 - (-4)$

$4 + 4$

$\boxed{8}$

9) $3 - 7y$, $y = 4$

$3 - 7(4)$

$3 - 28$

$\boxed{-25}$

10) $-\frac{z}{4} + 1$, $z = 3$

$-\frac{3}{4} + \frac{1}{1}$

$-\frac{3}{4} + \frac{4}{4}$

$\boxed{\frac{1}{4}}$

11) $\frac{5x + 1}{2 + 4x}$, $x = 0$

PEMDAS

$\frac{5(0) + 1}{2 + 4(0)}$

$\frac{0 + 1}{2 + 0}$

$\boxed{\frac{1}{2}}$

12) $5(x - 1)^2 + 11$, $x = -2$

$5(-2 - 1)^2 + 11$

$5(-3)^2 + 11$

$5(9) + 11$

$45 + 11$

$\boxed{56}$

13) $7 - 4y^2$, $y = 1$

$7 - 4(1)^2$

$7 - 4(1)$

$7 - 4$

$\boxed{3}$

(14) $4-(1-z)^2$, $z=-4$

$4-(1-(-4))^2$

$4-(1+4)^2$

$4-5^2$

$4-25$

$\boxed{-21}$

(15) a) x^2, $x=-7$

$(-7)^2$

$\boxed{49}$

b) $-y^2$, $y=-9$

$-(-9)^2$

$\boxed{-81}$

c) $-z^2$, $z=2$

$-1z^2$

$-1(2)^2$

$-1(4)=\boxed{-4}$

d) $-x^3$, $x=-2$

$-1x^3$

$-1(-2)^3$

$-1(-8)=\boxed{8}$

(16) a) $3z^2$, $z=-2$

$3(-2)^2$

$3(4)$

$\boxed{12}$

b) $-4x^2$, $x=-3$

$-4(-3)^2$

$-4(9)$

$\boxed{-36}$

(17) a) $-4y^3$, $y=-2$

$-4(-2)^3$

$-4(-8)$

$\boxed{32}$

b) $9z^3$, $z=-1$

$9(-1)^3$

$9(-1)$

$\boxed{-9}$

(18) $2x^3-4(x+1)+2$, $x=-2$

$2(-2)^3-4(-2+1)+2$

$2(-2)^3-4(-1)+2$

$2(-8)-4(-1)+2$

$-16-(-4)+2$

$-16+4+2$

$\boxed{-10}$

(19) $7+\frac{3y-1}{2}+2(3y+1)^2$, $y=0$

$7+\frac{3(0)-1}{2}+2(3(0)+1)^2$

$7+\frac{0-1}{2}+2(0+1)^2$

$7-\frac{1}{2}+2(1)^2$

$7-\frac{1}{2}+2(1)$

$7-\frac{1}{2}+2$

$\frac{9}{1}-\frac{1}{2}=\frac{18}{2}-\frac{1}{2}=\boxed{\frac{17}{2}}$

(20) a) Find E if $E=mc^2$, $m=4$ and $c=5$.

$E=4(5)^2=4(25)=\boxed{100}$

b) Find y if $y=ax^2+bx+c$, $a=3$, $b=-2$, $c=1$, and $x=2$.

$y=3(2)^2+(-2)(2)+1=3(4)+(-2)(2)+1$

$=12-4+1$

$=\boxed{9}$

(21) a) Find A if $A=\frac{h(b_1+b_2)}{2}$, $h=1$, $b_1=2$, and $b_2=0$.

$A=\frac{(1)(2+0)}{2}=\frac{(1)(2)}{2}=\frac{2}{2}=\boxed{1}$

b) Find A if $A = 2WH + 2LH + 2HW$, $H=4$, $L=3$, and $W=2$.

$$A = 2(2)(4) + 2(3)4 + 2(4)(2)$$

$$A = 16 + 24 + 16$$

$$A = \boxed{56}$$

Check the solutions below. If correct, write "True." If incorrect, write "False." Show the work that leads to your answer.

22 a) $3x + 1 = -2x - 9$, $x = -2$

$$3(-2)+1 = -2(-2)-9$$

$$-6+1 = 4-9$$

$$\boxed{-5 = -5}$$

$\boxed{\text{True}}$

b) $7y + 8y - 7 = 4 - 3y - 8y$, $y = 2$

$$7(2)+8(2)-7 = 4-3(2)-8(2)$$

$$14+16-7 = 4-6-16$$

$$\boxed{23 \neq -18}$$

$\boxed{\text{False}}$

23 a) $3z - z + 1 = -2z + 5$, $z = -1$

$$3(-1)-(-1)+1 = -2(-1)+5$$

$$-3+1+1 = 2+5$$

$$\boxed{-1 \neq 7}$$

$\boxed{\text{False}}$

b) $x - 10x + 9 = 2x - 8x + 5$, $x = 0$

$$0-10(0)+9 = 2(0)-8(0)+5$$

$$0-0+9 = 0-0+5$$

$$\boxed{9 \neq 5}$$

$\boxed{\text{False}}$

Solve.

24 $$5(z+2)-(3z-4) = 2z - 4(1-2z)+10$$

$$5z+10-3z+4 = 2z-4+8z+10$$

$$2z+14 = 10z+6$$

$$8 = 8z$$

$$\boxed{1 = z}$$

25 $$-3(y+2)+2(3y+3)+(-3) = 10-2(y-4)$$

$$-3y-6+6y+6+(-3) = 10-2y+8$$

$$3y-3 = 18-2y$$

$$5y = 21$$

$$\boxed{y = \frac{21}{5}}$$